"PhD candidate for interpretable machine learning in medical image analysis with industry experience in automotive and medical domains. I am excited to contribute to the future of machine intelligence and looking forward to the next challenge."

COSMIN I. **BERCEA**

PhD Candidate
Interpretable Machine Learning

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Languages

Romanian: native
German: native
English: fluent

WORK EXPERIENCE

Doctoral Researcher - Helmholtz Center Munich / TUM
2020 - Present - Munich, Germany
Prof. Dr. Daniel Rückert, Prof. Dr. Julia Schnabel

- Interpretable machine learning in medical image analysis

Research Engineer CV/ML - Bosch Corporate Research
2018 - 2020 (2 Years) - Hildesheim, Germany
Dr.-Ing. Niklas Beuter

- Solved challenging perception and behaviour prediction tasks in the field of scene understanding and interior monitoring

Research Intern - Siemens Healthineers
2017 - 2018 (1 Year | 7 Months) - Erlangen, Germany
Dr.rer.nat. Olivier Pauly, Dr.-Ing. Florin C. Ghesu

- Improved accuracy and robustness of lung segmentation on chest X-rays by reducing the number of failures by 50%
- Developed novel recurrent neural networks that share a memory module

Research Intern - Computer Vision Center
2016 (6 Months) - Barcelona, Spain
Prof. Dr. Petia Radeva

- Developed a temporal deep learning model for ego-centric action recognition

Research Intern - FAU Pattern Recognition Lab
2015-2016 (1 Year) - Erlangen, Germany
Dr.-Ing. Thomas Köhler, Prof. Dr.-Ing. habil. Andreas Maier

- Increased state-of-the-art image super-resolution by 3.0 dB by implementing a novel, robust multi-frame super-resolution method

Research Assistant - Fraunhofer IIS
2013-2017 (2 Years | 10 Months) - Nuremberg, Germany
Dr.-Ing. Christopher Mutschler

- Implemented cheap and efficient camera-based indoor localisation
- Assured QoS in a complex event processing system (C++)

EDUCATION

TUM: Technical University of Munich
2020 - (Dr. rer.nat. Computer Science) - Munich, Germany
Prof. Dr. Daniel Rückert, Prof. Dr. Julia Schnabel

- Interpretable machine learning in medical image analysis

FAU: University Erlangen-Nuremberg
Prof. Dr. Andreas Maier
2011 - 2018 (B/M.Sc., Computer Science)
- Grade 1.6, Erlangen, Germany

- Pattern recognition and medical image analysis

UAB: Universitat Autonoma de Barcelona
2016 (Exchange Semester) - Barcelona, Spain

- Deep learning and computer vision

PUBLICATIONS & PATENTS

- **Cosmin I. Bercea,** Daniel Rueckert, and Julia A. Schnabel. *"What do we learn? Debunking the Myth of Unsupervised Outlier Detection."* arXiv preprint arXiv:2206.03698 (2022).
- **Cosmin I. Bercea**, Olivier Pauly, Andreas K. Maier, Florin C. Ghesu: *SHAMANN: Shared Memory Augmented Neural Networks,* IPMI, Honk Kong, 2019
- **Cosmin Bercea**, A. Maier, T. Köhler: *Confidence-aware Levenberg-Marquardt optimization for joint motion estimation and super-resolution,* ICIP, Pheonix, 2016
- **Cosmin I. Bercea**, Alexander Lengsfeld, Lucas Rego Drumond: *Memory Networks for Detecting Anomalies in Vehicle Occupant Behaviour,* 2019
- Alexander Lengsfeld, Philipp Lenz, Lucas Rego Drumond, **Cosmin I. Bercea:** *Pedestrian Behaviour Recognition for Improving the AEB functionality*, 2019
- **Cosmin I. Bercea**, Florin C. Ghesu, Olivier Pauly: *Memory Networks for Medical Image Analysis,* 2018